Society for Computer Technology & Research's

PUNE INSTITUTE OF COMPUTER TECHNOLOGY

Web Technology Lab [310257]

Mini-Project Report on

[AI-Powered Personal Finance Tracker]

Submitted by

[Kunal Bagul, Roll No. 31403]

[Swarupanand Deshmukh, Roll No. 31411]

[Prathamesh Ghadge, Roll No. 31415]

Class: - T.E (2019 course) Branch: -Computer Engineering

ACADEMIC YEAR: 2025-26

TABLE OF CONTENTS

CHAPTER 1
INTRODUCTION

The **AI-Powered Personal Finance Tracker** is a web-based application developed to help individuals efficiently manage their financial activities in a digital environment. In today's modern lifestyle, tracking income and expenses has become essential for maintaining financial stability. This system provides users with a secure platform where they can register, log in, and manage their financial transactions easily. By organizing financial data into structured records, the application simplifies financial management and improves overall accuracy compared to traditional manual methods.

The system allows users to record their income and expenses in categorized formats, making it easier to monitor spending habits. It also includes an interactive dashboard that displays financial summaries such as total income, total expenses, and current balance using visual charts and graphs. These visual elements help users quickly understand their financial condition without manually calculating totals. The system stores all transaction details in a database, ensuring that financial information remains secure and accessible at any time.

Some key features introduced in this system include:

- Secure **user registration and login functionality**
- Efficient **management of income and expense transactions**
- Interactive **dashboard with charts and financial summaries**
- Reliable **database storage** for maintaining financial records

Overall, this project demonstrates how modern web technologies can be used to build a practical financial management solution that improves accuracy, usability, and decision-making.

CHAPTER 2
MOTIVATION

The motivation behind developing the **AI-Powered Personal Finance Tracker** comes from the challenges faced by individuals while managing their finances using traditional methods. Many people still rely on notebooks or spreadsheets to record their daily expenses and income. These manual methods are often time-consuming, difficult to maintain, and prone to human errors such as missing entries or incorrect calculations. As the number of financial transactions increases, keeping track of them manually becomes inefficient and unreliable.

Another important factor motivating this project is the need for better financial awareness and visualization. Without proper tools, users find it difficult to analyze their spending habits and identify areas where money is being overspent. The use of dashboards and charts in this system helps users visualize financial data clearly, making it easier to monitor trends and make informed financial decisions. Additionally, advancements in web technologies and database systems made it possible to create a digital solution that is secure, efficient, and accessible from anywhere.

The major reasons that motivated the development of this project include:

- Reducing **manual effort and calculation errors** in financial tracking
- Providing **real-time insights** into income and expenses
- Improving **financial awareness** through visual representations
- Utilizing **modern technologies** to create a reliable financial tool

Thus, the motivation for this project lies in solving real-world financial tracking problems by providing a smarter and more efficient digital solution.

CHAPTER 3
OBJECTIVE

The main objective of the **AI-Powered Personal Finance Tracker** is to develop a secure and user-friendly system that enables users to manage their financial transactions digitally. The system aims to replace traditional manual record-keeping methods with an automated solution that improves accuracy and efficiency. By allowing users to store and organize their financial data in a structured manner, the system helps in maintaining better control over personal finances.

Another important objective of this project is to provide full transaction management capabilities through CRUD (Create, Read, Update, Delete) operations. Users can add new income or expense records, update incorrect entries, view transaction history, and delete unnecessary records. In addition to transaction management, the system also focuses on data visualization through dashboards and graphical charts. These features help users analyze their financial patterns and make informed decisions regarding spending and saving.

The major objectives of the system include:

- Developing a **secure authentication system** for user access
- Enabling **CRUD operations** for managing financial transactions
- Providing **visual dashboards** for financial data analysis
- Designing a **reliable backend and database system**
- Creating a **responsive and user-friendly interface**

Overall, the objective of this project is to build a comprehensive financial tracking system that improves financial organization, enhances usability, and supports better financial decision-making.

CHAPTER 4

SCOPE OF PROJECT

This project mainly focuses on providing users with a digital platform to record, manage, and analyze their financial transactions in an organized manner. The system includes essential features such as user authentication, transaction management, and dashboard visualization, which together help users monitor their income and expenses efficiently. By clearly defining the scope, the project ensures that the system remains focused on its primary goal of simplifying personal financial management.

Within the defined scope, the system supports several core operations that allow users to interact with their financial data easily. Users can securely register and log in to their accounts, add new transactions, and manage existing records. The dashboard provides a clear overview of financial information through summaries and graphical charts. Additionally, all user and transaction data is stored in a database, ensuring that records remain available across multiple sessions. These features make the system reliable and useful for day-to-day financial tracking.

The scope of the project mainly includes the following functionalities:

- **User Management:** Registration, login, and logout functionalities for secure access
- **Transaction Management:** Adding, viewing, updating, and deleting income and expense records
- **Dashboard Analytics:** Displaying summaries of total income, expenses, and balance
- **Data Visualization:** Showing financial data through charts and graphs
- **Database Storage:** Maintaining user and transaction data securely for future access

CHAPTER 5

INTENDED AUDIENCE

The **AI-Powered Personal Finance Tracker** is designed for individuals who want an efficient and organized way to manage their personal finances. The primary audience includes students, working professionals, and small business users who regularly handle income and expenses. Many individuals face difficulties in tracking daily spending, saving money, and understanding their financial habits. This system provides a simple and user-friendly platform that allows users to record financial transactions and analyze their financial condition without requiring advanced technical knowledge.

The intended audience of this system includes:

- **Students** who want to manage pocket money and track daily expenses
- **Working professionals** who need to monitor salary income and monthly spending
- **Small business owners** who want to maintain simple financial records
- **Individuals with limited technical knowledge** who require an easy-to-use financial tool
- **Anyone interested in improving financial discipline and saving habits**

Overall, the project is intended for a wide range of users who want a simple, reliable, and accessible solution for managing their personal financial data.

CHAPTER 6

FUNCTIONAL REQUIREMENTS

Functional requirements describe the main operations and services that the system must perform to meet user needs. In the **AI-Powered Personal Finance Tracker**, functional requirements focus on managing users, recording financial transactions, and displaying financial insights. These functions ensure that users can securely access the system, perform financial operations, and view meaningful information about their income and expenses.

The system includes several essential features that allow users to interact with their financial data efficiently. Each function is designed to simplify financial management and provide a seamless user experience. The following are the main functional requirements of the system:

- **User Registration:** Allows new users to create an account by entering username, email, and password.
- **User Login:** Enables registered users to securely log in using valid credentials.
- **User Logout:** Allows users to safely exit their session and protect account security.
- **Add Transaction:** Enables users to enter new income or expense details such as amount, category, and date.
- **View Transactions:** Displays a list of all recorded transactions for easy tracking.
- **Update Transaction:** Allows users to modify existing transaction details when needed.
- **Delete Transaction:** Enables removal of unwanted or incorrect transaction records.
- **Dashboard Display:** Shows financial summaries such as total income, total expenses, and net balance.
- **Chart Visualization:** Displays graphical representations of financial data to help users understand spending patterns.

CHAPTER 7
NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe how the system performs its functions rather than what functions it performs. These requirements focus on aspects such as performance, security, reliability, and usability. In the **AI-Powered Personal Finance Tracker**, non-functional requirements ensure that the system operates smoothly, securely, and efficiently while providing a good user experience.

Performance is an important factor in ensuring that the application responds quickly to user actions such as login, transaction entry, and dashboard loading. Security is also a critical requirement because financial data is sensitive and must be protected from unauthorized access. The system uses authentication mechanisms and secure data storage to maintain privacy and confidentiality. Additionally, reliability ensures that the system consistently saves user data without loss or corruption.

The major non-functional requirements of the system include:

- **Performance:** The system should load pages quickly and process transactions without delay.
- **Security:** User data must be protected through secure login mechanisms and encrypted passwords.
- **Usability:** The interface should be simple, intuitive, and easy to navigate for all users.
- **Reliability:** The system should store data accurately and prevent data loss.
- **Scalability:** The system should handle increasing numbers of users and transactions efficiently.
- **Availability:** The application should remain accessible whenever users need to manage their finances.

CHAPTER 8

OPERATIONAL ENVIRONMENT

The operating environment requirements for this project are listed below:

Database Requirements:

- **Neon Database**
- Used for storing user details and financial transaction records
- Supports secure and scalable cloud-based data storage

Software Requirements:

- **Operating System:** Windows, Linux, or macOS
- **Frontend Technologies:** HTML, CSS, JavaScript, React.js
- **Backend Technologies:** Node.js, Express.js
- **Web Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge
- **API Testing Tool (for development):** Postman
- **Version Control Tool:** Git and GitHub

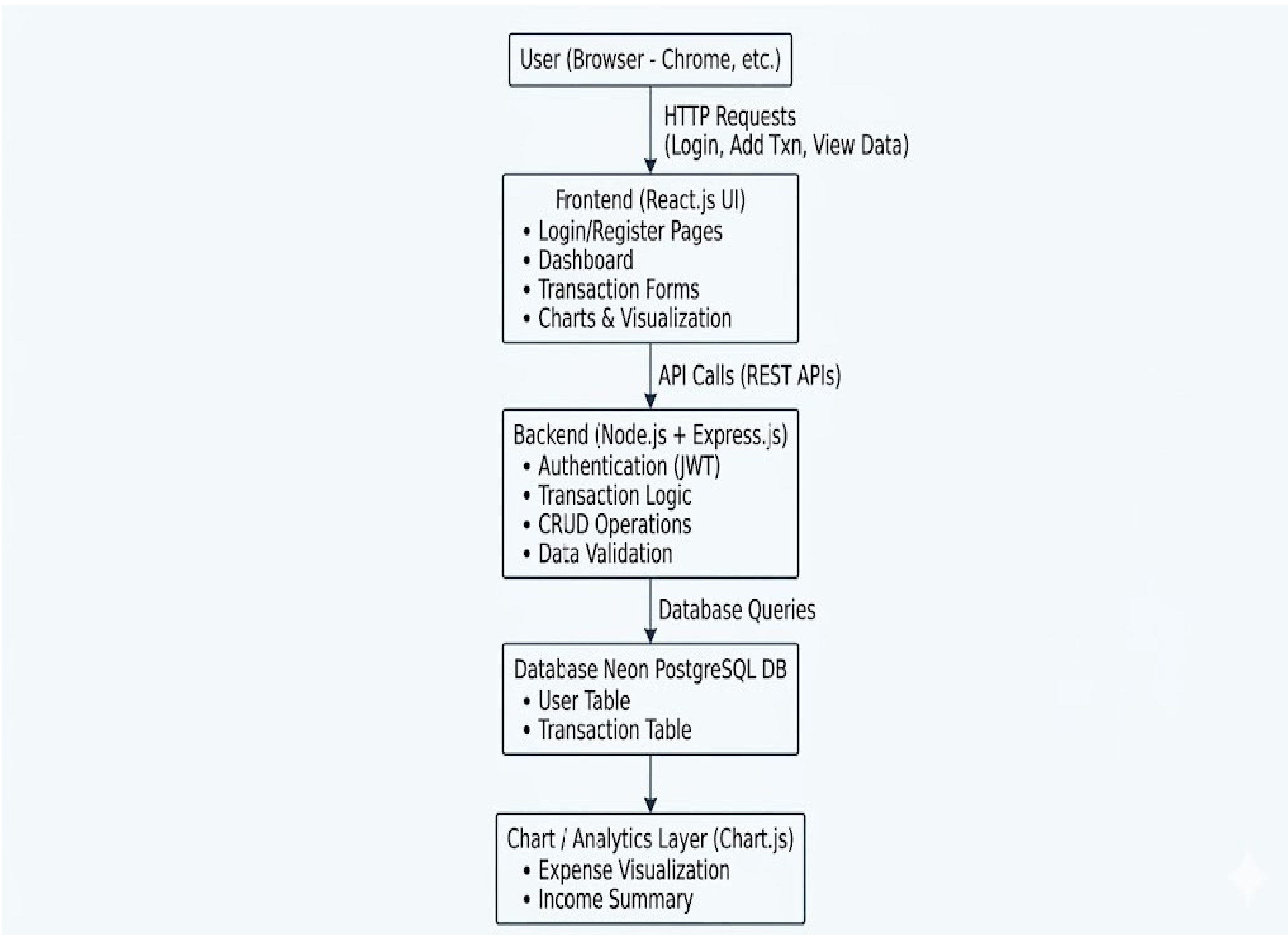
Hardware Requirements:

- **Processor:** 1 GHz or faster processor
- **RAM:** Minimum 1 GB
- **Storage:** Minimum 500 MB free disk space

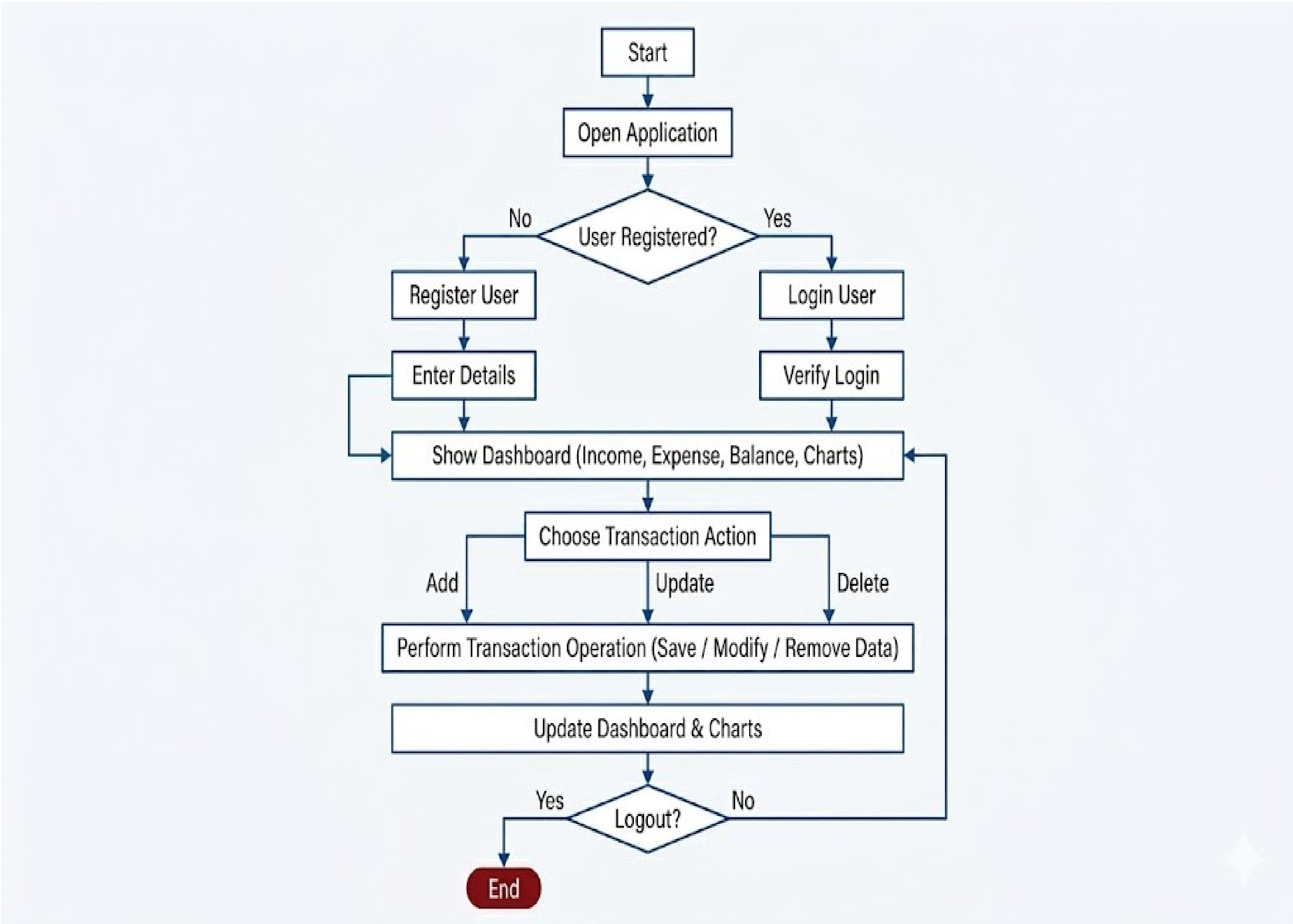
CHAPTER 9

SYSTEM ARCHITECTURE AND FLOWCHART

1. System Architecture:



2. Flowchart


Start
Open Application
User Registered?
No
Yes
Register User
Login User
Enter Details
Verify Login
Show Dashboard (Income, Expense, Balance, Charts)
Choose Transaction Action
Add
Update
Delete
Perform Transaction Operation (Save / Modify / Remove Data)
Update Dashboard & Charts
Yes
Logout?
No
End

CHAPTER 10

IMPLEMENTATION DETAILS

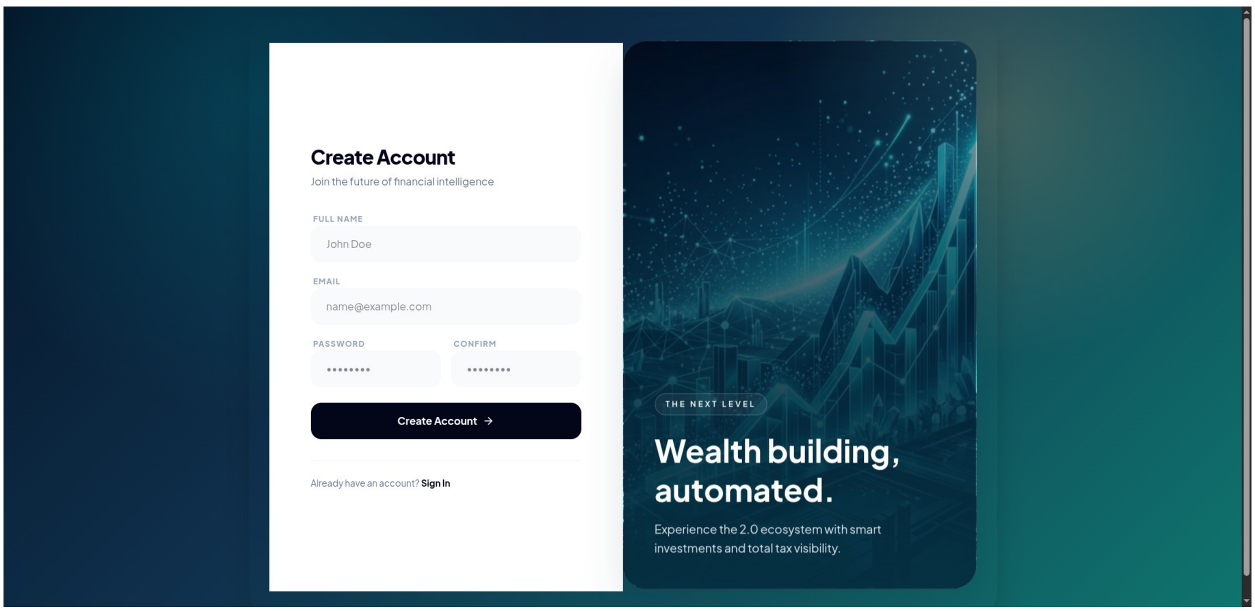
Frontend Link: https://github.com/Pratham-ghadge/ai-finance-tracker-frontend

Backend Link: https://github.com/Pratham-ghadge/ai-finance-tracker-backend

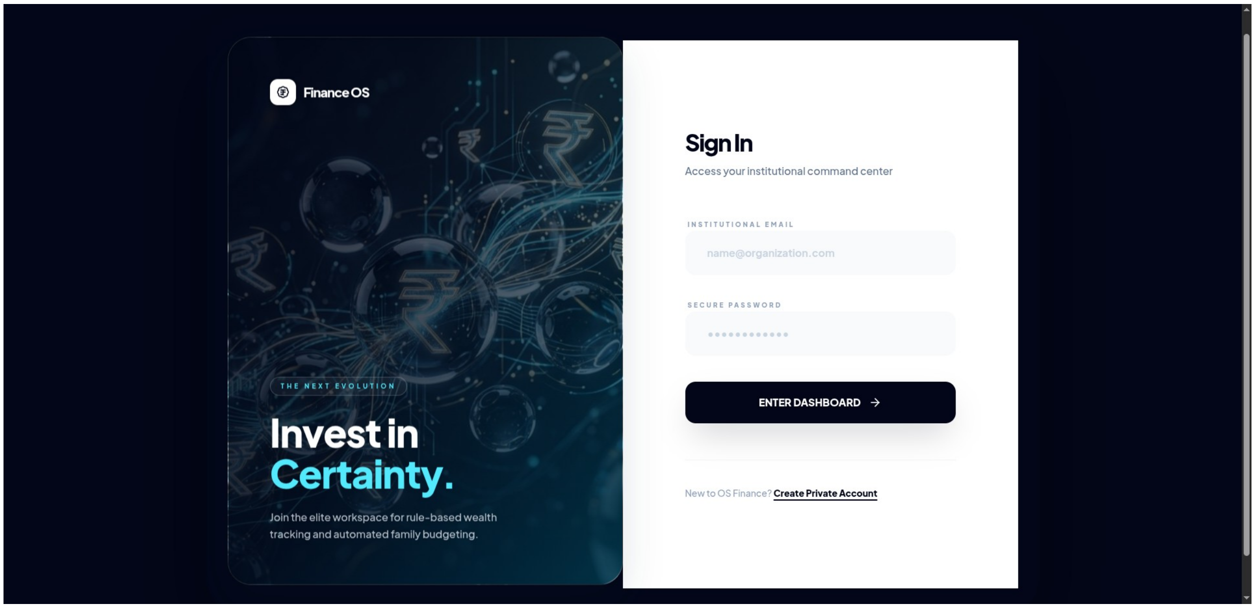
Demo Link: https://ai-finance-tracker-frontend-five.vercel.app/

SCREENSHOTS:

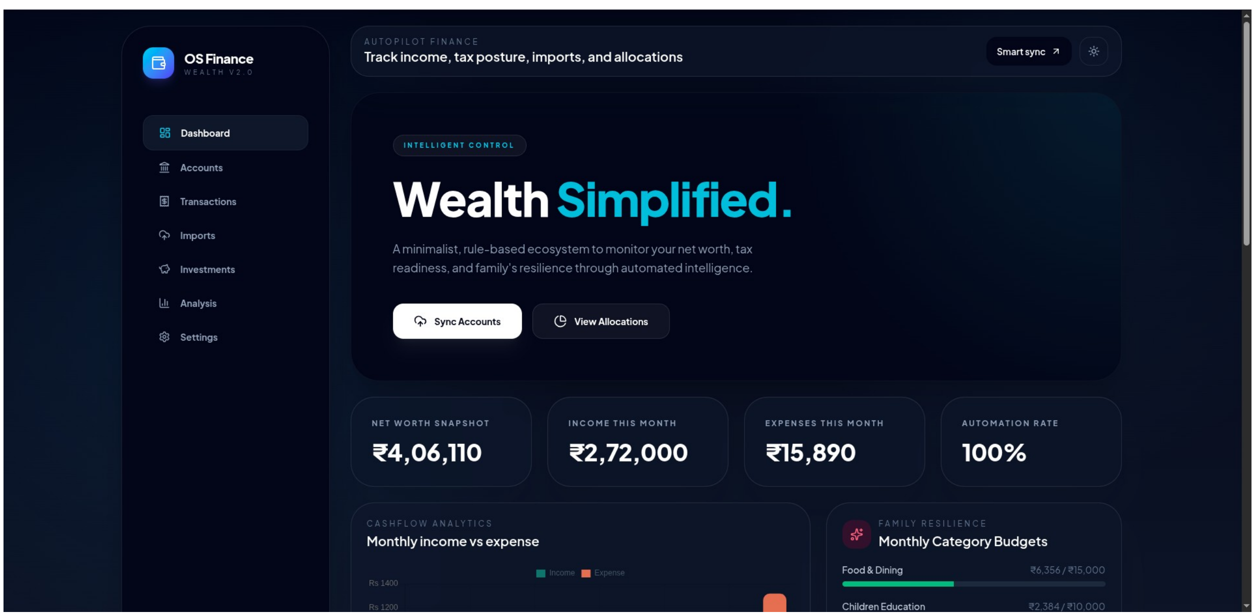
Sign Up:

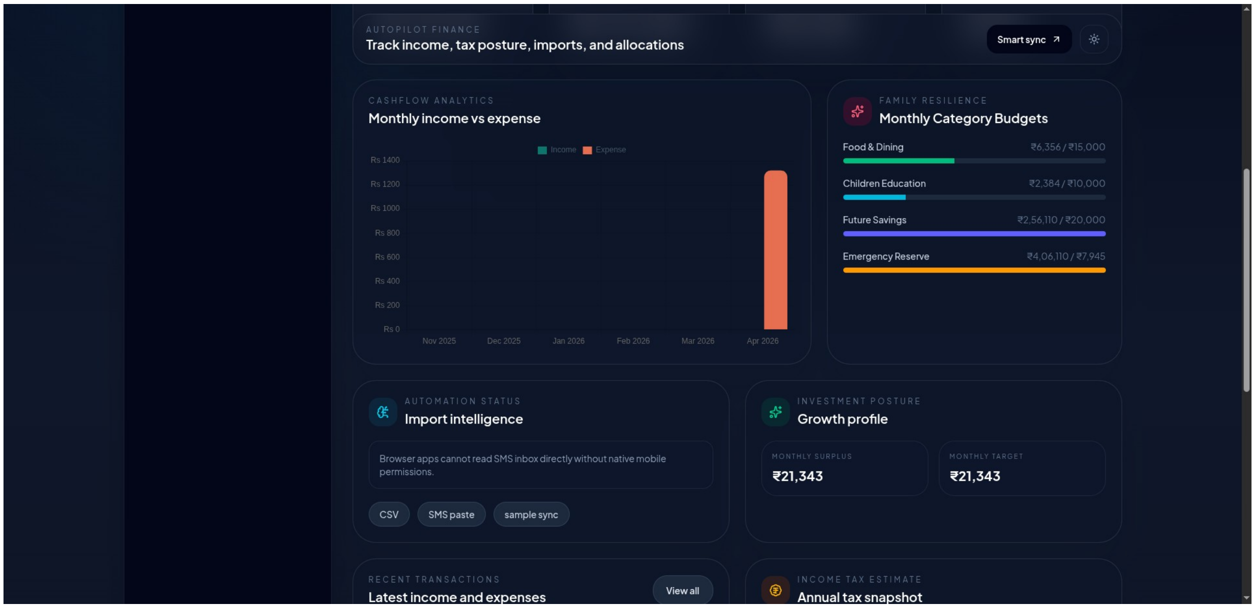


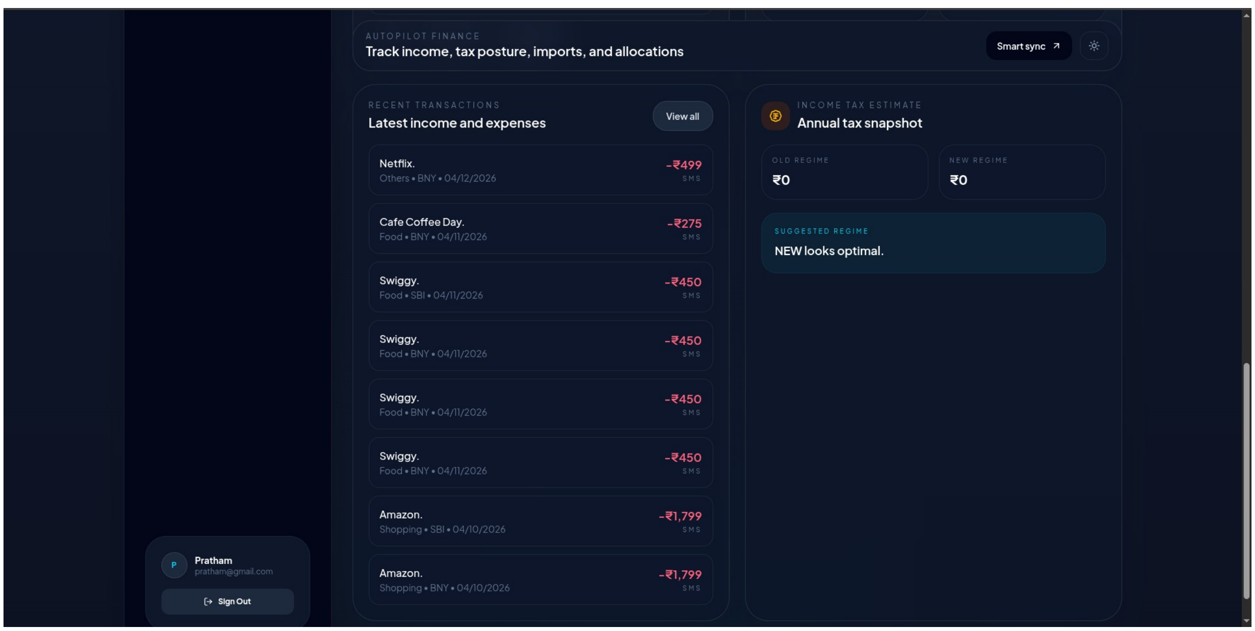
Sign in:



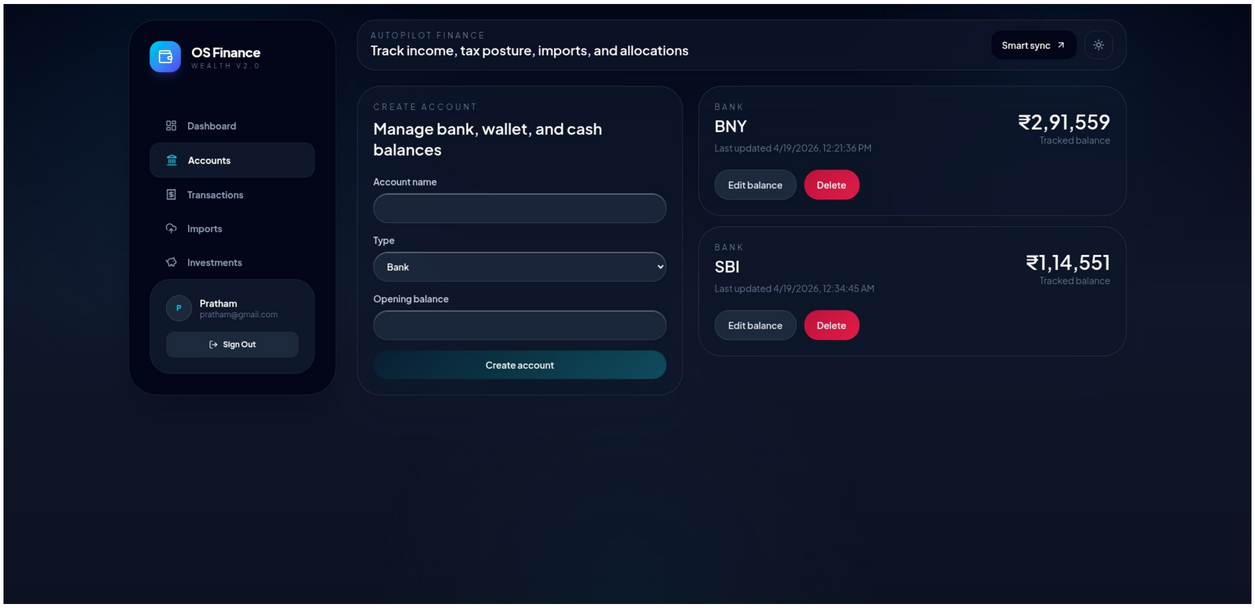
Dashboard:



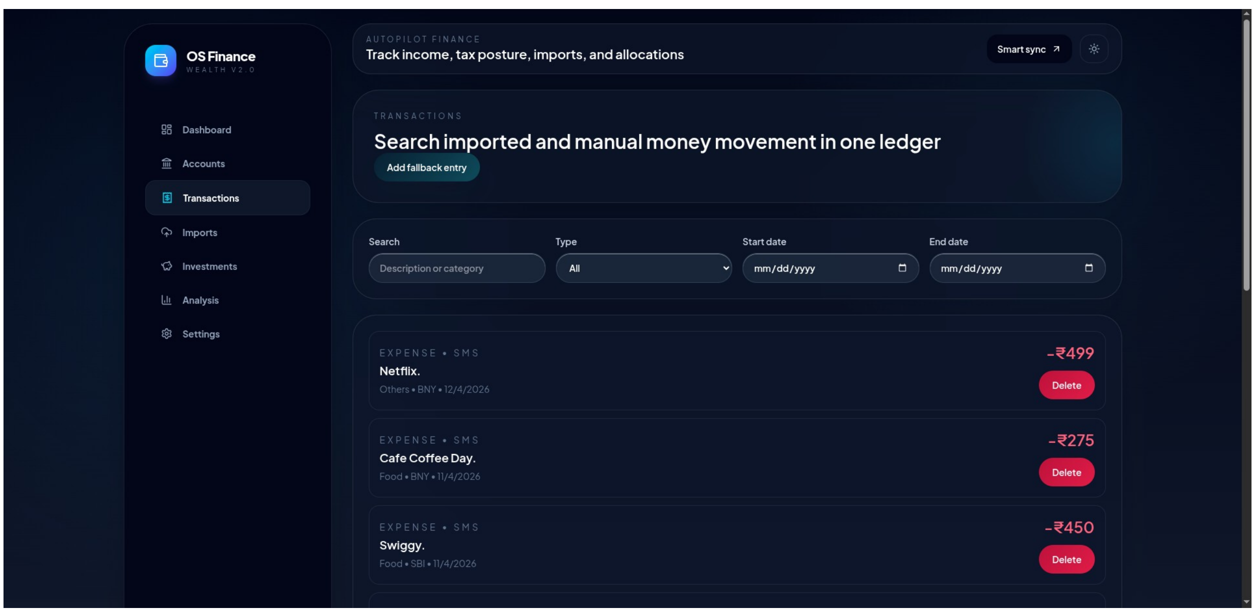

AUTOPILOT FINANCE
Track income, tax posture, imports, and allocations
Smart sync
CASHFLOW ANALYTICS
Monthly income vs expense
Income
Expense
Rs 1400
Rs 1200
Rs 1000
Rs 800
Rs 600
Rs 400
Rs 200
Rs 0
Nov 2025
Dec 2025
Jan 2026
Feb 2026
Mar 2026
Apr 2026
FAMILY RESILIENCE
Monthly Category Budgets
Food & Dining
₹6,356 / ₹15,000
Children Education
₹2,384 / ₹10,000
Future Savings
₹2,56,110 / ₹20,000
Emergency Reserve
₹4,06,110 / ₹7,945
AUTOMATION STATUS
Import intelligence
Browser apps cannot read SMS inbox directly without native mobile permissions.
CSV
SMS paste
sample sync
INVESTMENT POSTURE
Growth profile
MONTHLY SURPLUS
₹21,343
MONTHLY TARGET
₹21,343
RECENT TRANSACTIONS
Latest income and expenses
View all
INCOME TAX ESTIMATE
Annual tax snapshot


AUTOPILOT FINANCE
Track income, tax posture, imports, and allocations
Smart sync
RECENT TRANSACTIONS
Latest income and expenses
View all
Netflix.
Others • BNY • 04/12/2026
-₹499
SMS
Cafe Coffee Day.
Food • BNY • 04/11/2026
-₹275
SMS
Swiggy.
Food • SBI • 04/11/2026
-₹450
SMS
Swiggy.
Food • BNY • 04/11/2026
-₹450
SMS
Swiggy.
Food • BNY • 04/11/2026
-₹450
SMS
Swiggy.
Food • BNY • 04/11/2026
-₹450
SMS
Amazon.
Shopping • SBI • 04/10/2026
-₹1,799
SMS
Amazon.
Shopping • BNY • 04/10/2026
-₹1,799
SMS
INCOME TAX ESTIMATE
Annual tax snapshot
OLD REGIME
₹0
NEW REGIME
₹0
SUGGESTED REGIME
NEW looks optimal.
P
Pratham
pratham@gmail.com
Sign Out

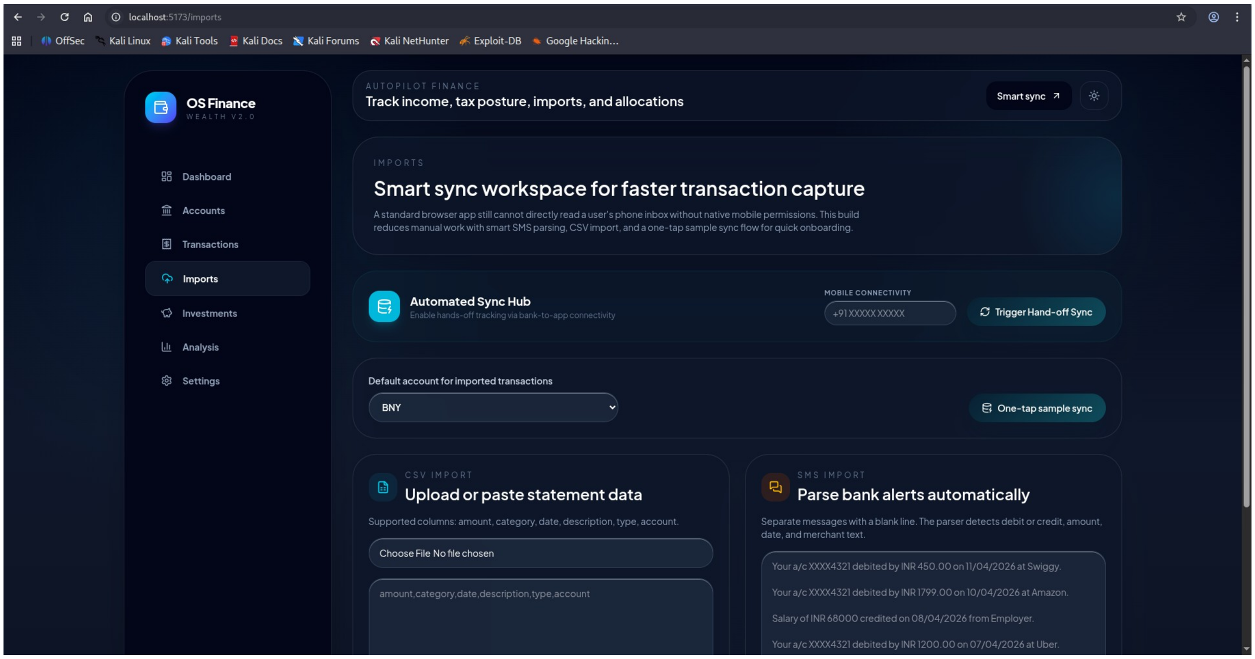
Accounts:

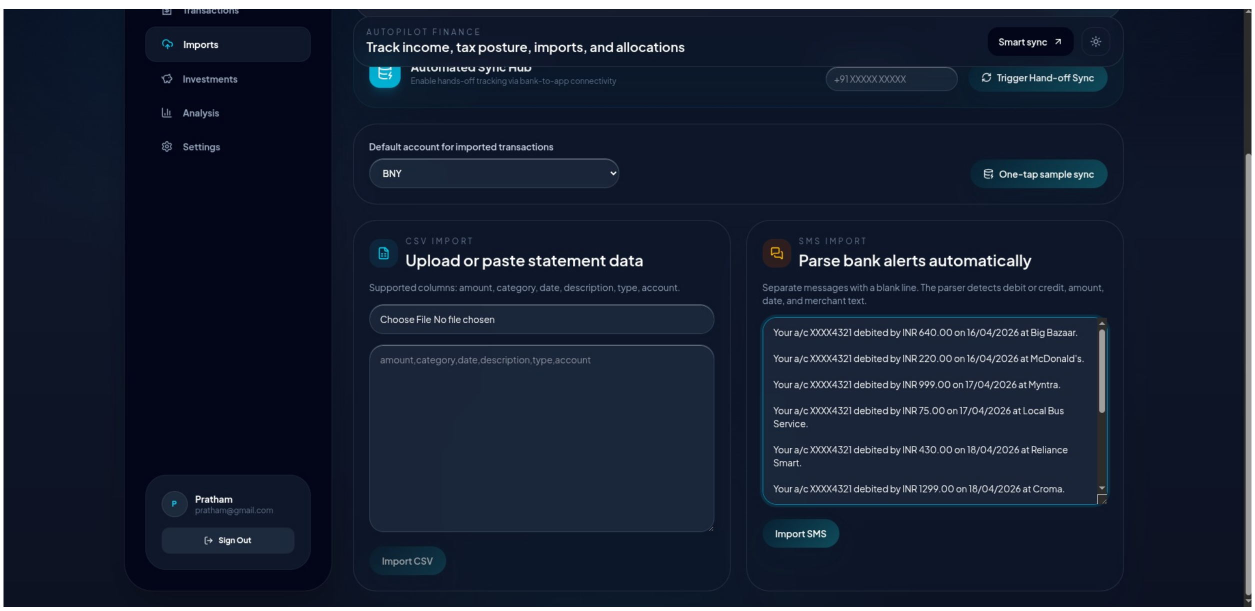


Transactions:

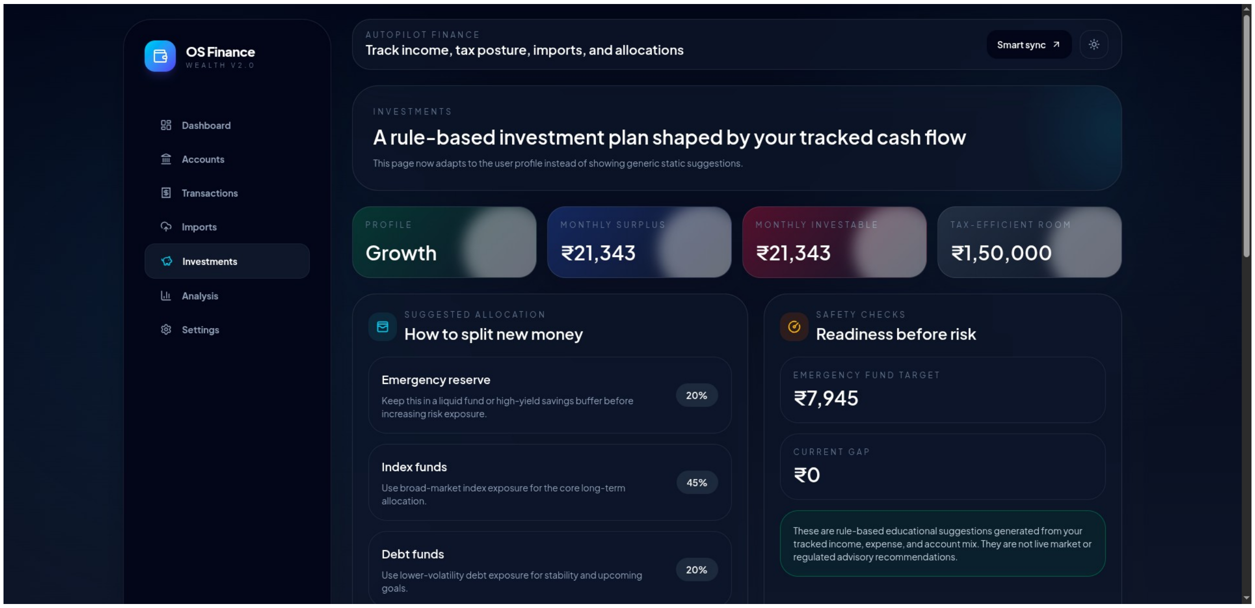


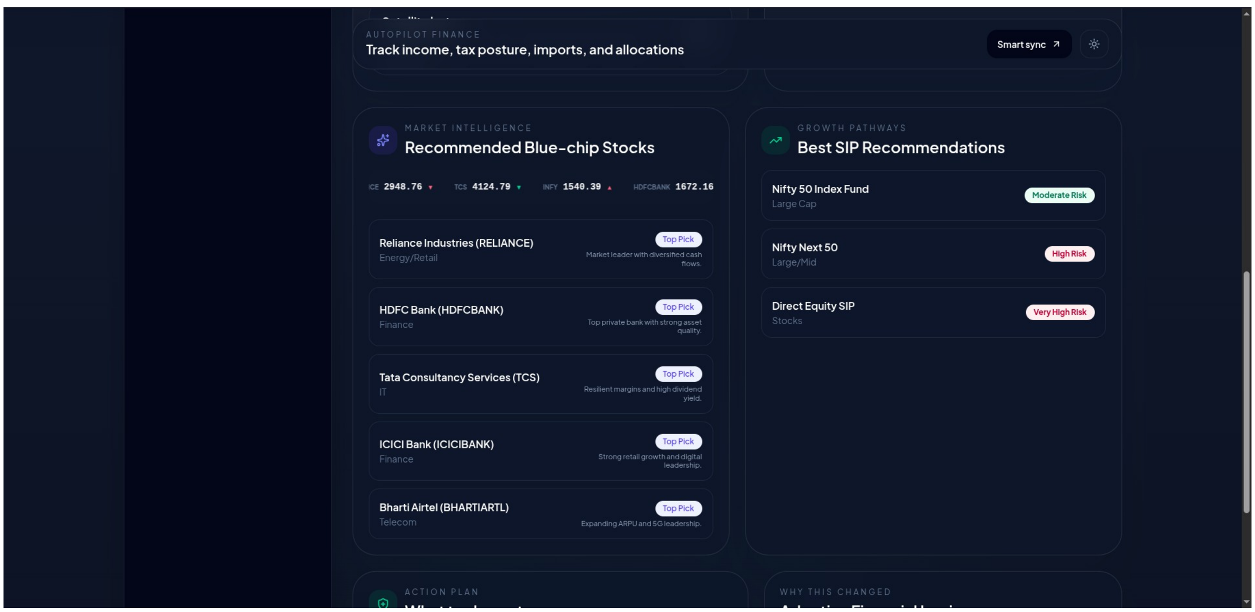
Imports:


localhost:5173/imports
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hackin...
OS Finance
WEALTH V2.0
Dashboard
Accounts
Transactions
Imports
Investments
Analysis
Settings
AUTOPILOT FINANCE
Track income, tax posture, imports, and allocations
Smart sync
IMPORTS
Smart sync workspace for faster transaction capture
A standard browser app still cannot directly read a user's phone inbox without native mobile permissions. This build reduces manual work with smart SMS parsing, CSV import, and a one-tap sample sync flow for quick onboarding.
Automated Sync Hub
Enable hands-off tracking via bank-to-app connectivity
MOBILE CONNECTIVITY
+91 XXXXX XXXXX
Trigger Hand-off Sync
Default account for imported transactions
BNY
One-tap sample sync
CSV IMPORT
Upload or paste statement data
Supported columns: amount, category, date, description, type, account.
Choose File No file chosen
amount,category,date,description,type,account
SMS IMPORT
Parse bank alerts automatically
Separate messages with a blank line. The parser detects debit or credit, amount, date, and merchant text.
Your a/c XXXX4321 debited by INR 450.00 on 11/04/2026 at Swiggy.
Your a/c XXXX4321 debited by INR 1799.00 on 10/04/2026 at Amazon.
Salary of INR 68000 credited on 08/04/2026 from Employer.
Your a/c XXXX4321 debited by INR 1200.00 on 07/04/2026 at Uber.


Imports
Investments
Analysis
Settings
AUTOPILOT FINANCE
Track income, tax posture, imports, and allocations
Smart sync
Enable hands-off tracking via bank-to-app connectivity
+91 XXXXX XXXXX
Trigger Hand-off Sync
Default account for imported transactions
BNY
One-tap sample sync
CSV IMPORT
Upload or paste statement data
Supported columns: amount, category, date, description, type, account.
Choose File No file chosen
amount,category,date,description,type,account
Import CSV
SMS IMPORT
Parse bank alerts automatically
Separate messages with a blank line. The parser detects debit or credit, amount, date, and merchant text.
Your a/c XXXX4321 debited by INR 640.00 on 16/04/2026 at Big Bazaar.
Your a/c XXXX4321 debited by INR 220.00 on 16/04/2026 at McDonald's.
Your a/c XXXX4321 debited by INR 999.00 on 17/04/2026 at Myntra.
Your a/c XXXX4321 debited by INR 75.00 on 17/04/2026 at Local Bus Service.
Your a/c XXXX4321 debited by INR 430.00 on 18/04/2026 at Reliance Smart.
Your a/c XXXX4321 debited by INR 1299.00 on 18/04/2026 at Croma.
Import SMS
Pratham
pratham@gmail.com
Sign Out

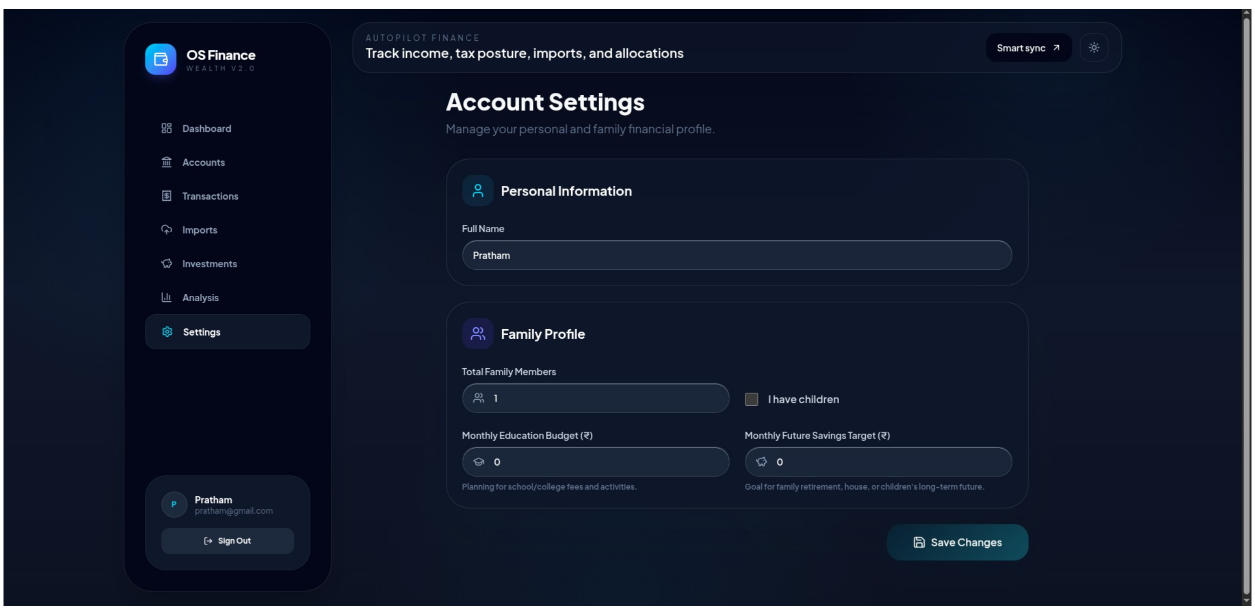
Investments:





Analysis:

Settings:



CHAPTER 11
CONCLUSION

The **AI-Powered Personal Finance Tracker** project was successfully developed to provide a simple and efficient solution for managing personal financial activities. The system enables users to securely register, log in, and manage their income and expense records in a structured digital format. By replacing traditional manual methods such as notebooks or spreadsheets, the project improves accuracy, reduces calculation errors, and saves time. The integration of database storage ensures that all financial records are safely stored and easily accessible whenever required.

The project also provides an interactive dashboard that displays financial summaries and visual charts, helping users better understand their financial habits. Features such as adding, updating, viewing, and deleting transactions allow users to maintain complete control over their financial data. The use of modern web technologies and a cloud-based database improves system reliability and accessibility, making it suitable for everyday financial tracking.

Overall, the project demonstrates how technology can be used to simplify financial management and improve financial awareness. With further enhancements such as budgeting tools, automated transaction integration, and advanced analytics, the system can be expanded into a more powerful and intelligent financial management platform in the future.

CHAPTER 12

REFERENCES

Official Documentation References:

- React.js Documentation
  https://react.dev/reference/react
- Node.js Documentation
  https://nodejs.org/en/docs
- Express.js Documentation
  https://expressjs.com/en/starter/installing.html
- Neon Database Documentation
  https://neon.tech/docs
- JSON Web Token (JWT) Documentation
  https://jwt.io/introduction
- Chart.js Documentation (for data visualization)
  https://www.chartjs.org/docs

Research Paper References:

- "A Study on Personal Finance Management Systems Using Web Applications"
  https://ieeexplore.ieee.org/document/8457935
- Design and Implementation of Money Management Web-Based Application (ScienceDirect)
  https://www.sciencedirect.com/science/article/pii/S187704281402000X
- Expense Tracker – An Expense Tracking Application Using OCR and Random Forest Algorithm (2025)
  https://www.researchgate.net/publication/395910560_EXPENSE_TRACKER-_AN_EXPENSE_TRACKING_APPLICATION_USING_OCR_AND_RANDOM_FOREST_ALGORITHM